



OOP FINAL PROJECT

B123040004 莊志文
B123040021 尤振德
B123040033 林雲翔



DIRECTORY



part 1

establish the environment

part 2

algorithm design

analysis & conclusion

part 3

motivation

game design

analysis & conclusion



PART 1

ESTABLISH THE ENVIRONMENT



PART 2

ALGORITHM DESIGN (based on Q-learning)

Learning Rate:

linear decay 、 dynamic increase upon positive outcomes

Reward & Penalty:

incentivizes shorter paths 、 optimized reward-to-penalty ration

Exploration Rate:

exponential decay over time 、 set a lower bound

ANALYSIS & CONCLUSIOIN

- Hybrid learning rate stabilizes convergence and strongly reinforces successful experience
- Adjusting the reward mechanism encourages the agent to select a safer path, balancing efficiency with risk avoidance
- exploration rate slowly decay ensure to cover the whole map, and lower bound avoid local optimal stagnation



PART 3

MOTIVATION

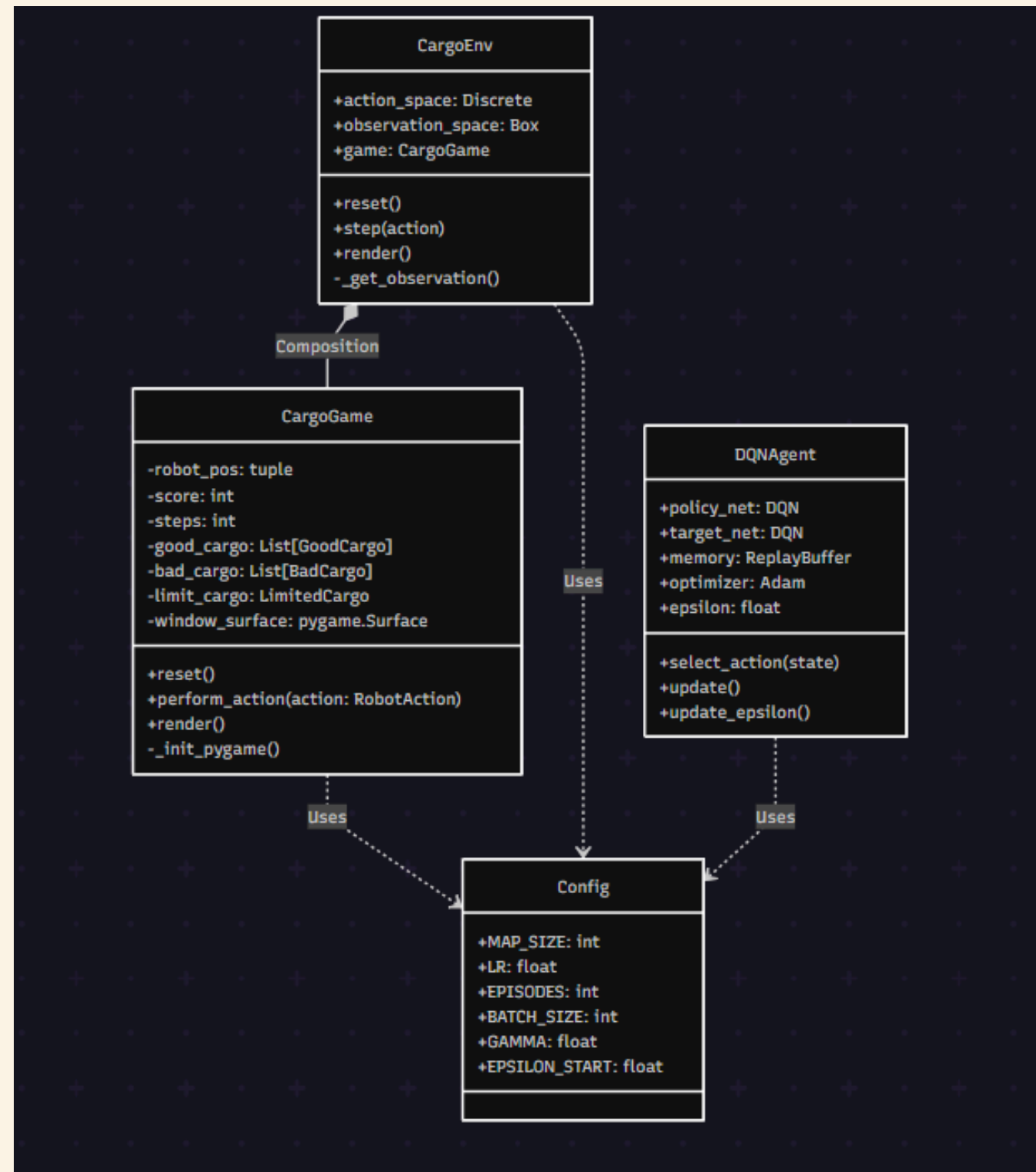
We thought that warehouse robot could be developed as a game.

So we added a concept about different cargos would be distributed in the map and let AI tried to get the highest score.

GAME DESIGN

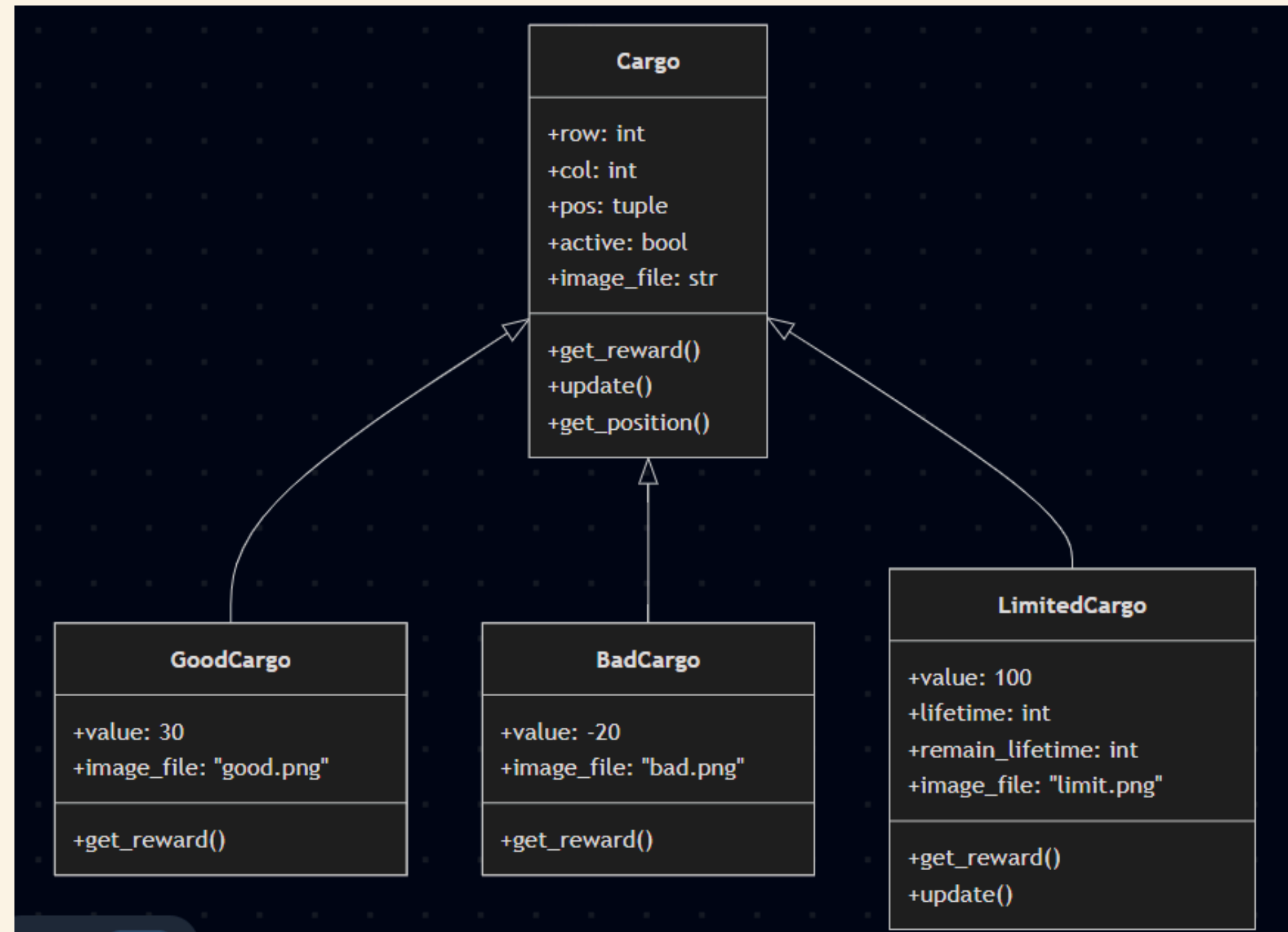
CONFIG.PY :

DEFINE THE ESSENTIAL
GAME SETTING AND THE
HYPERPARAMETERS OF
REINFORCEMENT LEARNING



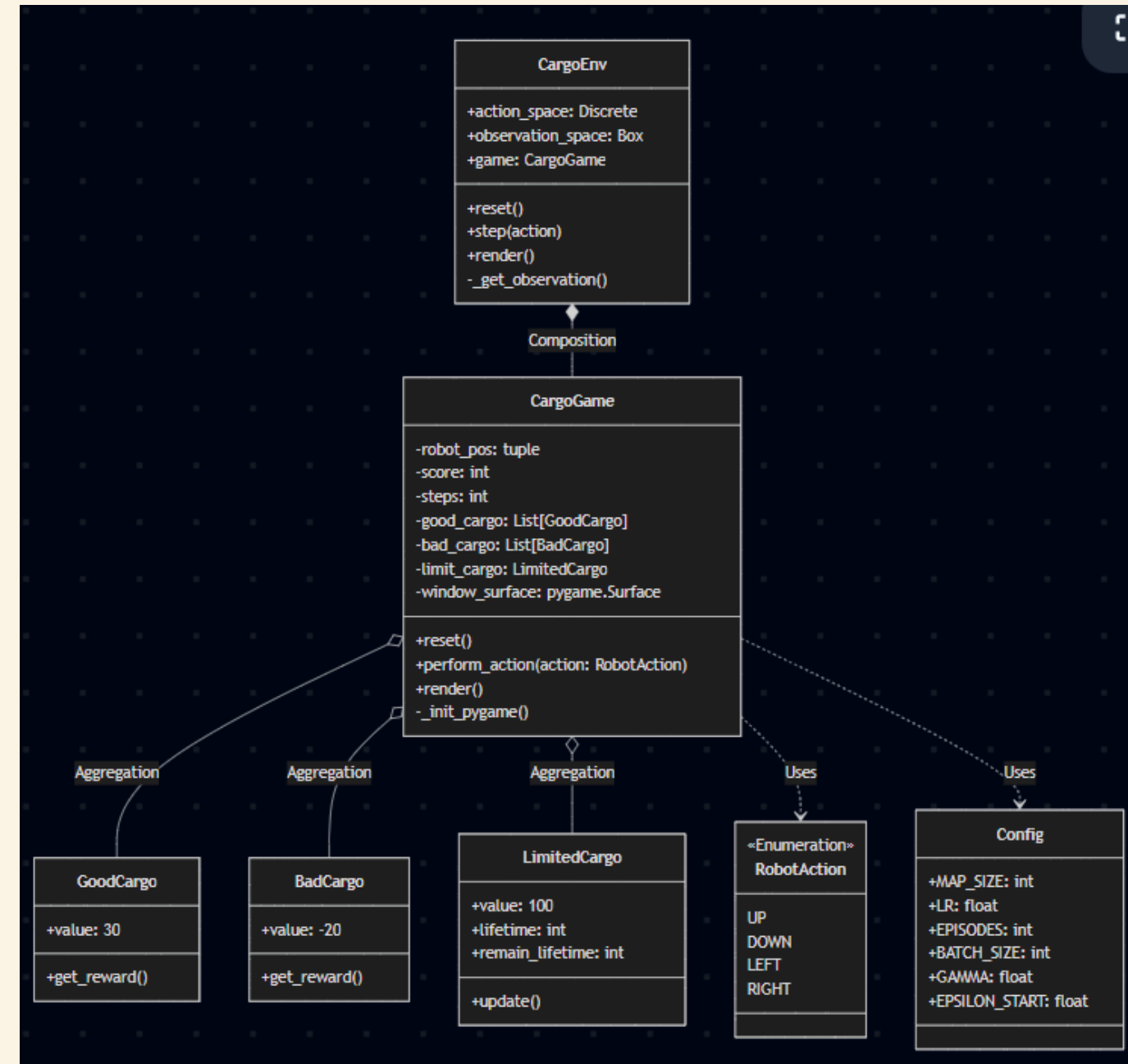
GAME DESIGN

CARGO.PY :
DEFINE THE FUELD AND
METHODS OF DIFFERENT
CARGOES



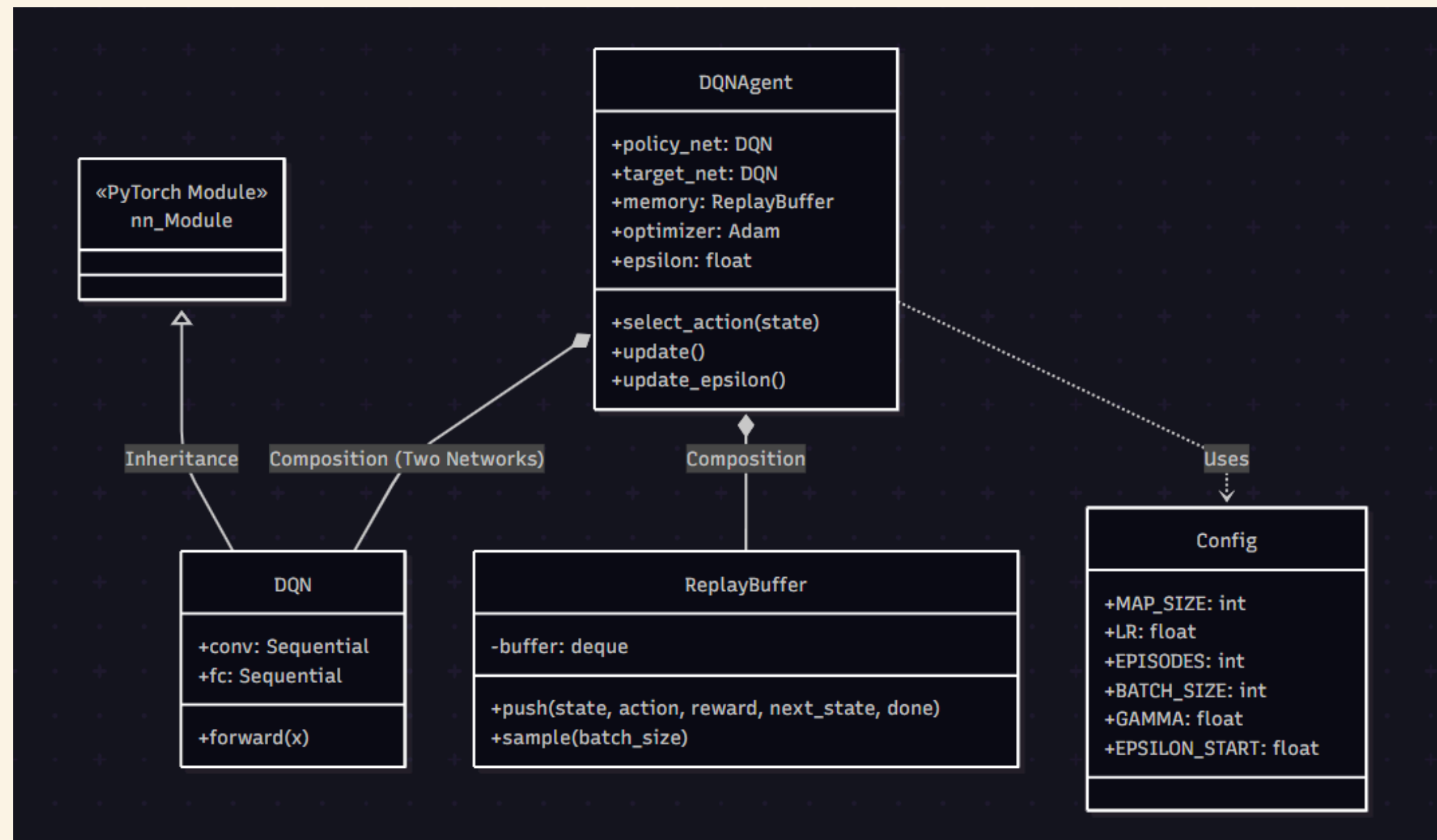
GAME DESIGN

ROBOT.PY :
SIMULATE THE ACTION
AND PASS THE RESULT TO
ENVIRONMENT.



GAME DESIGN

DDQN_AGENT.PY :
PREDICTS FUTURE REWARDS
TO MAKE PRECISE DECISIONS
FOR THE NEXT STEP.



GAME DESIGN

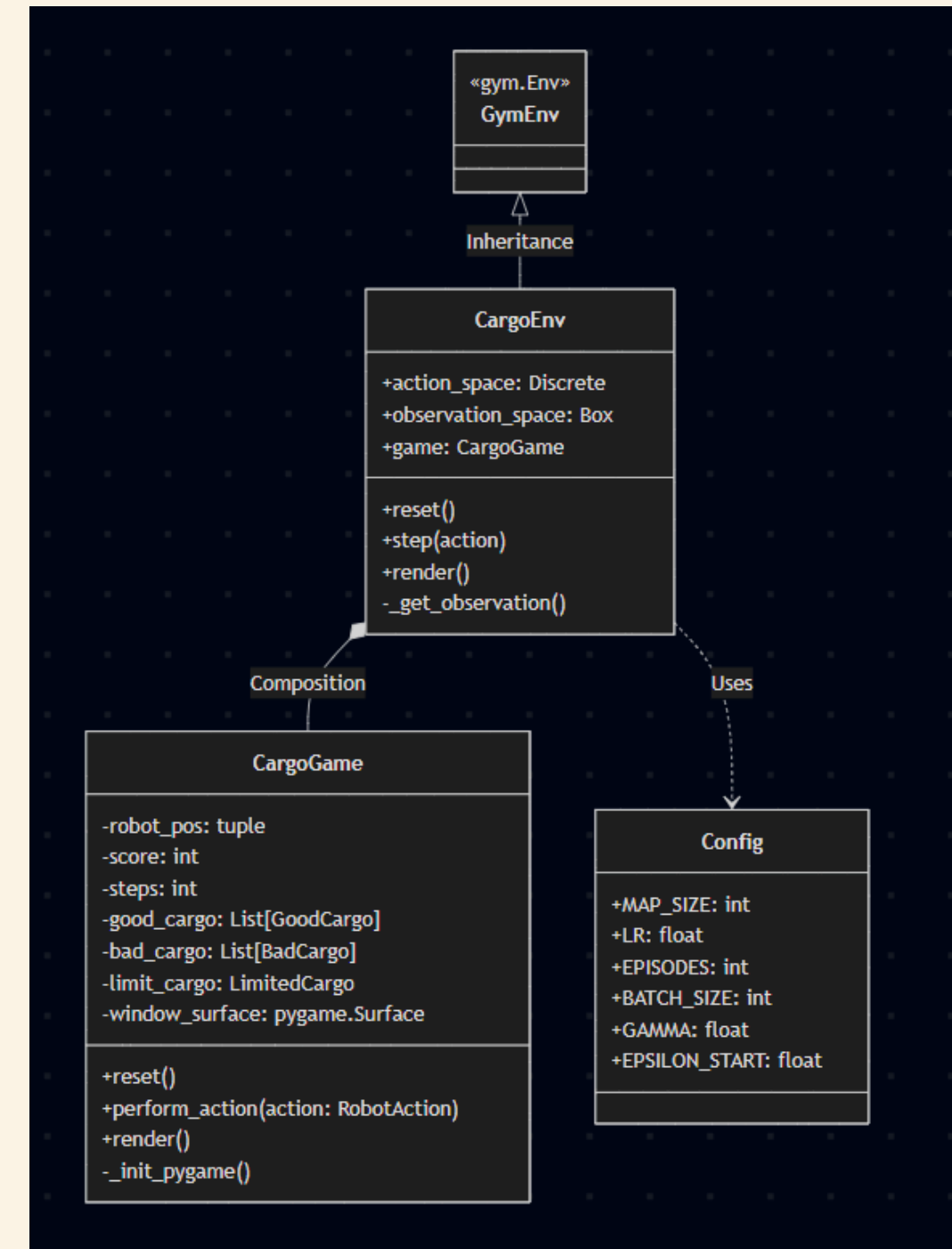
DDQN_AGENT.PY :
USING DOUBLE DQN
INSTEAD OF DQN

$$Y_{DDQN} = R + \gamma \cdot Q'(S', \arg \max_a Q(S', a; \theta); \theta')$$

$$L(\theta) = \mathbb{E} \left[(Y_{DDQN} - Q(S, A; \theta))^2 \right]$$

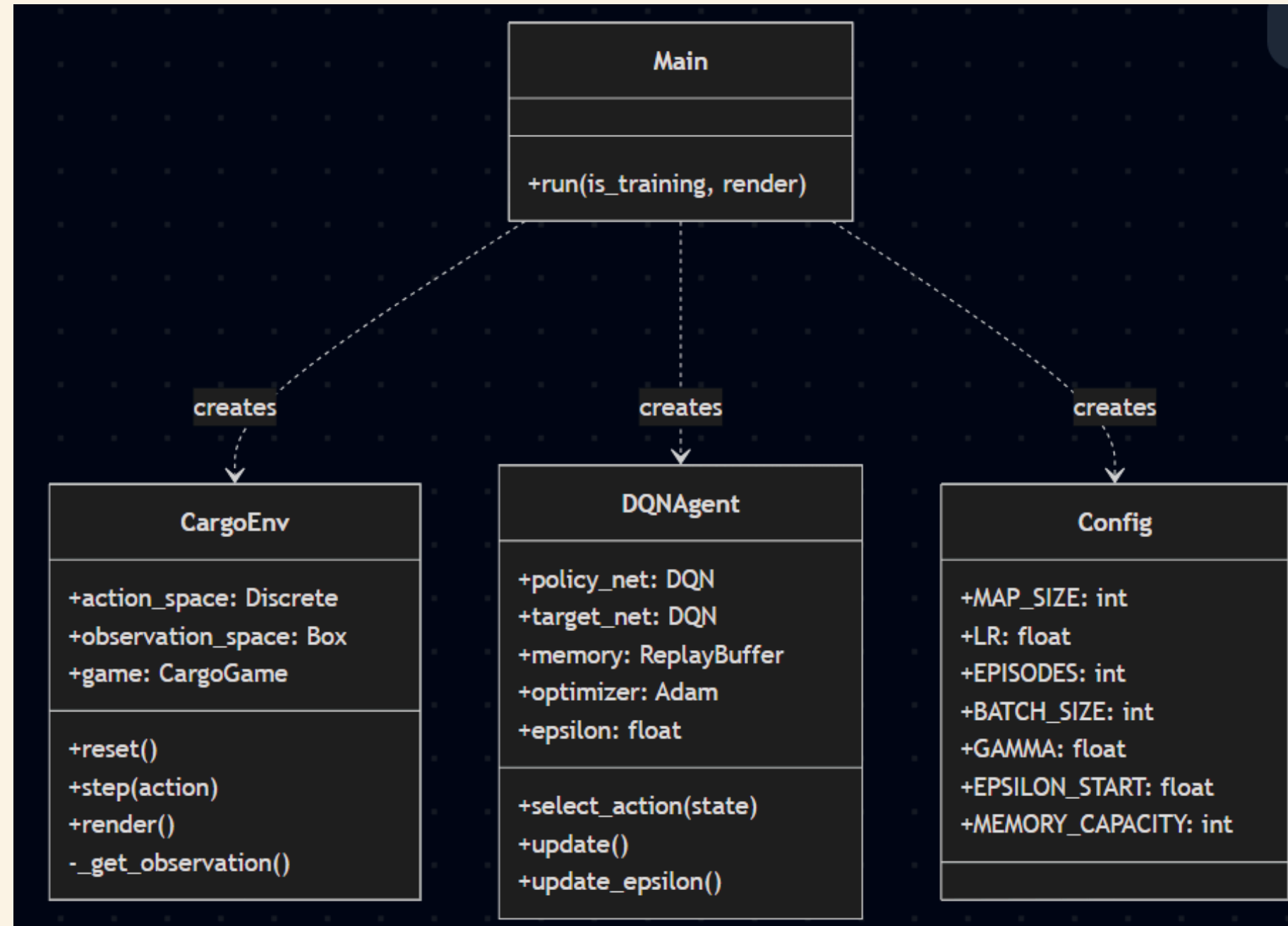
GAME DESIGN

ENVIRONMENT.PY :
THE INTERFACE OF THE
AGENT AND THE GAME.

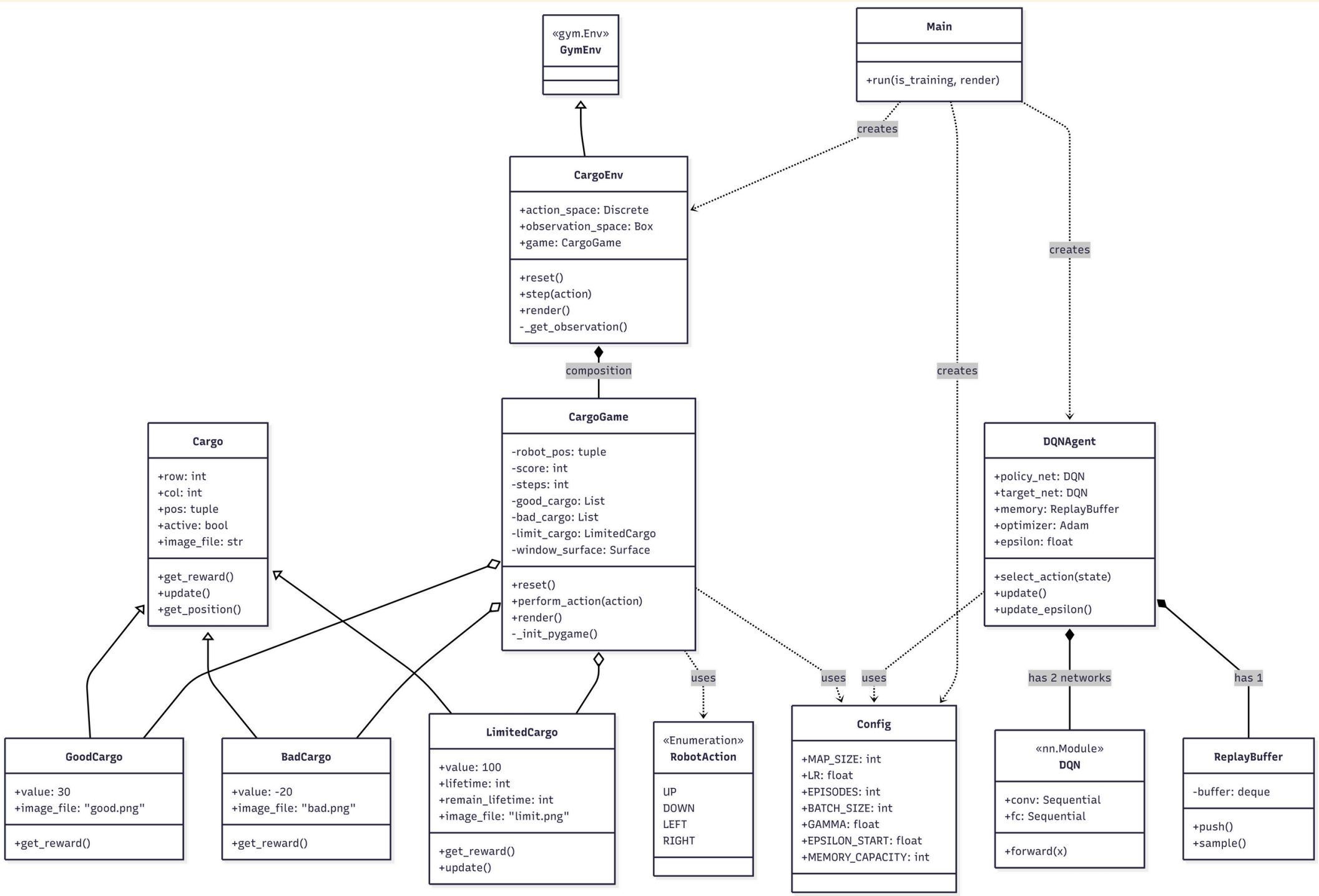


GAME DESIGN

MAIN.PY :
IMPLEMENT THE TRAINING
AND TESTING ENVIRONMENT

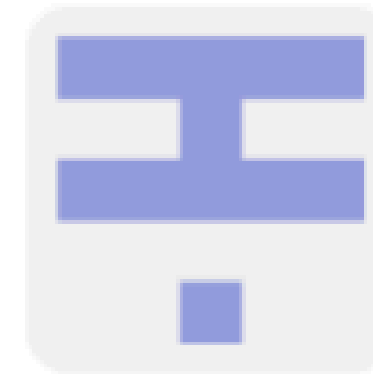


THE UML DIAGRAM



OUR GITHUB WEBSITE

Kevin-trytry/**Group-Project-main**



Contributors



Issues



Stars



Forks



Kevin-trytry/Group-Project-main

Contribute to Kevin-trytry/Group-Project-main development by creating an account on GitHub.



GitHub

Thank for listening!
